

Progress Report 2

Measuring the Recall Cost of Approximate Nearest Neighbor Search: A Parameter Selection Heuristic for Vector Databases

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1. Project Overview

The project measures how much recall approximate nearest neighbor (ANN) indexes silently give up at their library default settings, and tests whether the query setting a dataset needs can be predicted from a cheap statistic of the data instead of found by an exhaustive sweep. Progress Report 1 built the measurement harness and ran the first study on three datasets. In weeks 5 and 6 I moved the research contribution forward from the original weeks 9 to 11: I built the parameter-selection heuristic and evaluated it on datasets it had not seen. To make that evaluation meaningful I doubled the real datasets from three to six, added six synthetic datasets with a known intrinsic dimension, ran the result-set-size and full-collection checks promised last time, and put confidence intervals on the headline numbers. Two of those checks changed conclusions from the first report, and I document both below.

2. Work Completed

The codebase grew from 889 to 1,701 lines of Python. This period produced 2,556 new timed measurement records (1,332 from fine-grained sweeps, 636 from the Progress Report 1 grid on the new datasets, 450 from the result-set-size passes and 138 from full-size collections), bringing the total to 3,213. Every table and figure below regenerates from the committed CSVs with one script.

2.1 Six real datasets and six synthetic ones

I added Fashion-MNIST and MNIST (60,000 vectors, 784 dimensions, euclidean) and GloVe-25 (1,183,514 word vectors, 25 dimensions, angular) from ANN-Benchmarks, chosen because they differ from the first three in dimension, domain and size. I also wrote a generator for synthetic datasets with a controlled intrinsic dimension of 4, 8, 16, 24, 32 or 48: latent points from a 20-cluster Gaussian mixture pushed through a fixed random two-layer tanh network into 128 dimensions. Everything except the latent dimension is held constant, which isolates the one property the heuristic depends on.

2.2 Checking the dimensionality estimator against known answers

Because the synthetic intrinsic dimension is known, the Levina-Bickel estimator can be checked directly (Table 2). It recovers low dimensions well, 4.3 for a true 4 and 7.6 for a true 8, but underestimates increasingly as the true dimension grows, reading 22.8 for a true 48. This is the known downward bias of nearest-neighbor estimators with finite samples. The estimate stays monotone, which is what a ranking feature needs, but it means the synthetic sets only reach an LID of about 26.

2.3 Fine-grained sweeps for the heuristic

The heuristic needs, for each dataset, the smallest query setting that reaches a target recall. The coarse grid from the first report (ef doubling from 10 to 512) is too blunt for that, so I added a dense profile: 21 ef values from 10 to 1,024 on the default HNSW build, and 16 nprobe values from 1 to 256 on an IVF-Flat index with 1,024 lists, across all twelve datasets with three timed repeats each.

2.4 The heuristic and a held-out evaluation

The model is deliberately simple: $\log_2$ of the needed setting is a linear function of one dataset feature. I evaluated it with leave-one-dataset-out on the six real datasets. For each one, the model is fitted on the other five real datasets plus the six synthetic ones, predicts the held-out dataset's setting, the prediction is rounded up to the next measured grid value, and I read off the recall and latency that setting actually produced. It is compared against the library default, the oracle (cheapest setting that works, found by sweeping), a safe constant (the largest oracle value seen in training) and a median constant. A margin variant inflates the prediction by one root-mean-square training residual; I fixed that margin before looking at held-out results rather than tuning it.

2.5 Result-set size, full collections and confidence intervals

I ran the default HNSW and IVF-Flat builds at $k = 1$ and $k = 100$ on the first three datasets, and the default configurations on the full 1,000,000-vector SIFT and 1,183,514-vector GloVe-100 collections. The sweep now records the spread of recall across queries, so each recall carries a 95% interval. For the first report's defaults I rebuilt the same indexes single-threaded, which reproduces them exactly, to recover their intervals.

2.6 Two live demonstrations

`scripts/live_demo.py` shows the default gap on a fresh index in two seconds. `scripts/predict_demo.py` generates a dataset the model has never seen (intrinsic dimension 20 and a new seed), measures its LID in under a second, predicts nprobe, and checks the prediction live. On my run it measured an LID of 17.1, predicted nprobe 19, and reached recall 0.984 against a 0.95 target at 1.51 times the latency of the true cheapest setting, where the default reached 0.335. That test set comes from the same generator family as the synthetic training data, so it is held out in dimension and seed but not in kind; the real-dataset evaluation in section 3.3 is the stronger test.

dataset	vectors used	dim	metric	intrinsic dim (MLE)	LID at queries
SIFT	200,000	128	euclidean	19.9	23.9
GloVe-100	200,000	100	angular	36.4	43.1
NYTimes	200,000	256	angular	46.0	53.5
Fashion-MNIST	60,000	784	euclidean	15.8	17.6
MNIST	60,000	784	euclidean	14.1	16.0
GloVe-25	200,000	25	angular	15.7	18.2

Table 1. The six real datasets, ordered as in the project, with the dataset properties used as heuristic features.

true intrinsic dim	4	8	16	24	32	48
MLE estimate	4.3	7.6	13.0	16.6	19.2	22.8
LID at queries	4.9	9.1	15.4	18.9	21.7	25.7

Table 2. Estimator check on the six synthetic datasets, where the true intrinsic dimension is known.

3. Evidence of Progress and Results

3.1 The default gap on six datasets, with confidence intervals

Across all eighteen dataset-and-index combinations the mean default recall@10 is 0.536, but the spread is wide, from 0.128 (IVF-PQ on GloVe-100) to 0.930 (HNSW on Fashion-MNIST). That overturns a claim from the first report: with three datasets no default reached 0.75, but HNSW defaults on MNIST and Fashion-MNIST reach 0.929 and 0.930. The HNSW default does follow LID, dropping from about 0.93 at LID 16 to 18 down to 0.515 and 0.679 at LID 43 and 53, but LID is not the whole story: GloVe-25 has almost the same LID as Fashion-MNIST and its default reaches only 0.763. The two MNIST sets are also the smallest (60,000 vectors against 200,000), and section 3.6 shows defaults get worse as collections grow, so collection size is a confound I have not yet separated from LID. The IVF-Flat finding does generalize: on every one of the six datasets a tuned configuration beats the default at the same latency, by between 0.171 and 0.291 recall.

dataset (LID)	HNSW default recall@10	IVF-Flat default recall@10	IVF-Flat best at same latency	free gain
MNIST (16)	0.929 ± 0.014	0.642 ± 0.032	0.817	+0.175
Fashion-MNIST (18)	0.930 ± 0.013	0.698 ± 0.032	0.870	+0.171
GloVe-25 (18)	0.763 ± 0.025	0.564 ± 0.035	0.811	+0.247
SIFT (24)	0.731 ± 0.024	0.528 ± 0.034	0.773	+0.245
GloVe-100 (43)	0.515 ± 0.038	0.501 ± 0.038	0.681	+0.180
NYTimes (53)	0.679 ± 0.044	0.482 ± 0.043	0.774	+0.291

Table 3. Library defaults on all six datasets at 200,000 base vectors (or the full set when smaller), $k = 10$, sorted by LID. Intervals are 95% over the 300 queries.

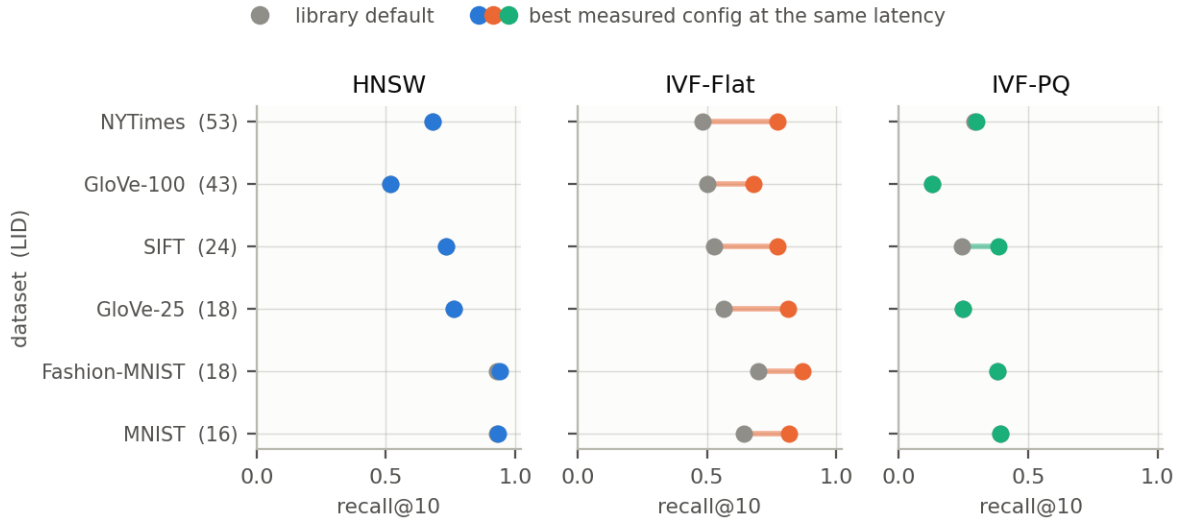
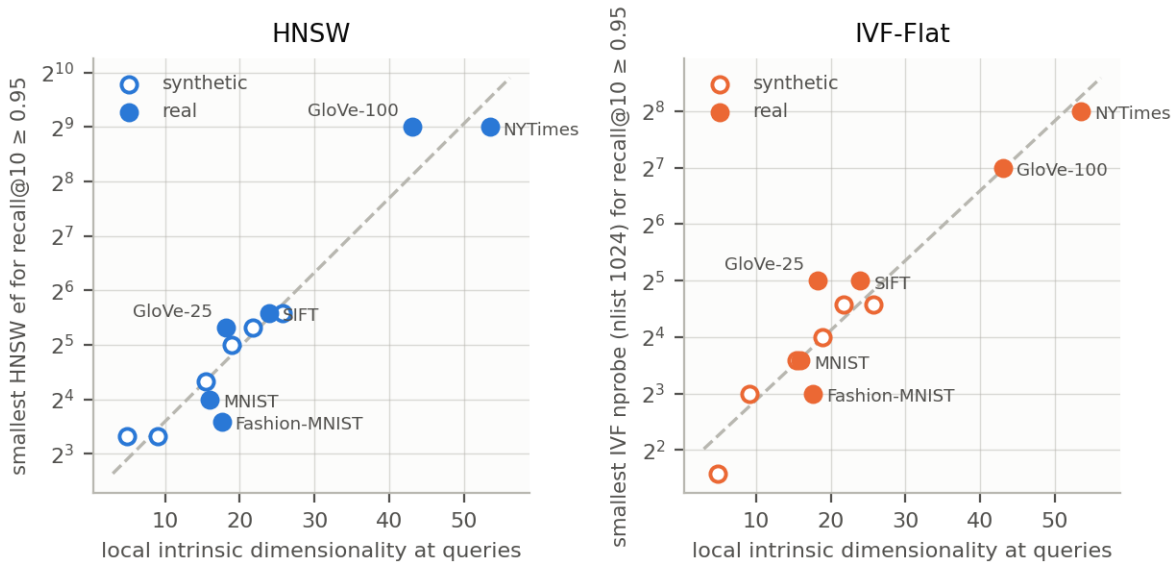


Figure 1. Default recall (gray) against the best measured configuration at the same latency (colored). Rows are sorted by LID; the number in parentheses is the LID.

3.2 The heuristic: LID predicts the setting a dataset needs

Across all twelve datasets, the log of the smallest setting that reaches recall 0.95 is almost linear in query-point LID, with a correlation of 0.963 for HNSW and 0.959 for IVF-Flat (Figure 2). The fitted HNSW model is $\log_2(\text{ef}) = 2.23 + 0.137 \times \text{LID}$, so the ef of a dataset needs doubles for every 7.3 points of LID; for IVF-Flat, $\log_2(\text{nprobe}) = 1.65 + 0.124 \times \text{LID}$, doubling every 8.1 points. The synthetic and real datasets fall on the same line, which is the result I was hoping for. The caveat is visible in the figure: synthetic points stop at LID 26, so the upper half of the line rests on just GloVe-100 and NYTimes.



3.3 Held-out evaluation

Table 4 is the core result of this period. The library default never reaches the 0.95 target on any held-out dataset. The safe constant always works for HNSW, but it pays 4.79 times the oracle's latency to do it, and still misses once on IVF-Flat. The median constant is cheap and misses four times out of six. The bare LID heuristic lands close to the right setting, at 1.17 and 1.00 times the oracle cost, and hits the target four times out of six; its misses are near-misses, with the lowest recall at 0.924 and 0.904. Adding the pre-registered margin raises that to five of six for both index families at 1.44 and 1.42 times the oracle cost, about a third of what the safe constant spends. Both remaining misses are GloVe word embeddings (GloVe-100 on HNSW reached 0.937, GloVe-25 on IVF-Flat 0.938), which suggests LID is missing something specific to that kind of data. At a target of 0.90 the margin variant hits all six IVF-Flat datasets at 1.51 times the oracle cost, where the safe constant hits five at 3.52 times.

method	HNSW hit rate	HNSW min recall	HNSW cost vs oracle	IVF hit rate	IVF min recall	IVF cost vs oracle
LID heuristic	4 / 6	0.924	1.17×	4 / 6	0.904	1.00×
LID heuristic + margin	5 / 6	0.937	1.44×	5 / 6	0.938	1.42×
safe constant	6 / 6	0.952	4.79×	5 / 6	0.908	4.68×
median constant	2 / 6	0.710	0.54×	2 / 6	0.770	0.50×
library default	0 / 6	0.503	0.22×	0 / 6	0.308	0.10×

Table 4. Leave-one-dataset-out results at a target of recall@10 = 0.95, over the six real datasets. Cost is the geometric mean of latency relative to the oracle setting.

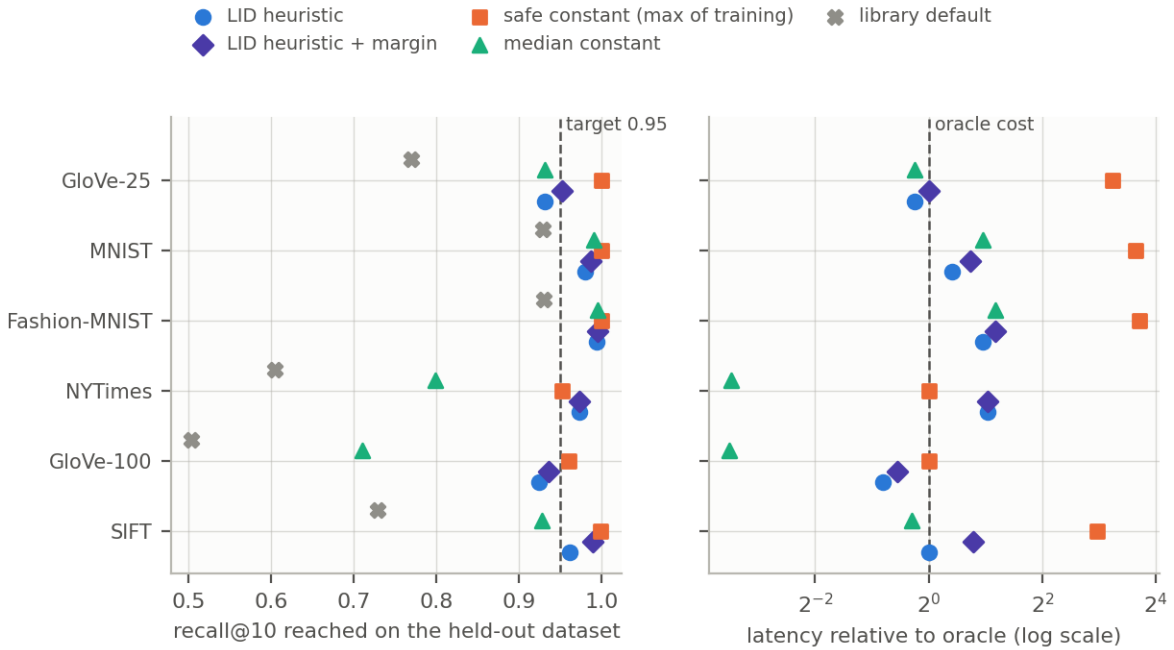


Figure 3. HNSW held-out detail. Left: recall reached on each held-out dataset. Right: latency relative to the oracle setting for that dataset (the default is omitted on the right because it never reaches the target).

3.4 Which feature carries the signal

To check that LID is doing the work, I refitted the same model with other features (Table 5). With LID, predictions are off by 0.72 log2 units on average for HNSW, about a factor of 1.6, and 0.47 units for IVF-Flat. Ambient dimension, the obvious naive choice, barely beats predicting a constant: 1.68 and 1.31 units against 1.97 and 1.54. The estimated intrinsic dimension is nearly as good as LID, which makes sense since both measure the same thing. I also refitted without the synthetic datasets. The effect is small and not uniform: HNSW error falls from 0.83 to 0.72 when synthetic data is added, IVF-Flat error from 0.50 to 0.47, but the bare IVF-Flat heuristic hits five of six with real data alone and four of six with synthetic data added. With six held-out datasets I cannot claim the synthetic data helps.

feature used by the model	HNSW mean $ \log_2 \text{error} $	IVF-Flat mean $ \log_2 \text{error} $
LID at queries	0.72	0.47
intrinsic dim (MLE)	0.72	0.57
ambient dimension	1.68	1.31
none (constant)	1.97	1.54

Table 5. Feature ablation, target 0.95. An error of 1.0 means the predicted setting was off by a factor of two on average.

3.5 Result-set size and the ef clamp

The first report inferred that hnsplib silently raises ef to at least k. This period I tested it directly: at $k = 100$, $ef = 10$ and $ef = 100$ give identical recall on all three datasets, to five decimals (0.94967, 0.73077 and 0.73317). So the documented default of $ef = 10$ quietly becomes $ef = 100$ whenever 100 neighbors are requested, which makes the default more accurate and about six times slower on SIFT (0.258 against 0.041 milliseconds per query). For a single nearest neighbor the default does better than at $k = 10$ (SIFT 0.790 against 0.729, GloVe-100 0.583 against 0.503, NYTimes 0.670 against 0.604), though the $k = 1$ intervals are wide, about ± 0.05 , because each query is a single hit or miss. The IVF-Flat default gets worse as k grows, from 0.543 at $k = 1$ to 0.458 at $k = 100$ on SIFT.

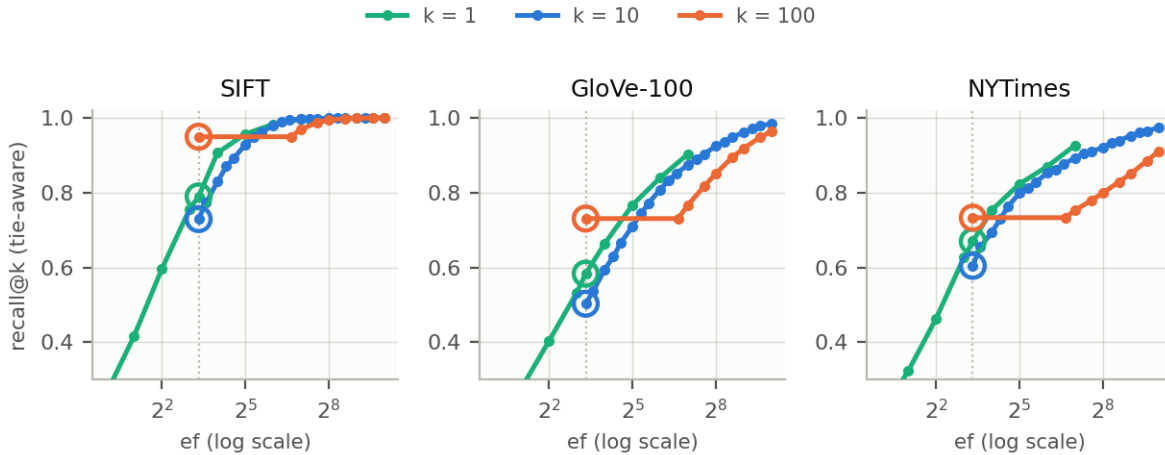


Figure 4. HNSW recall against ef at $k = 1, 10$ and 100 , default build. Rings mark the default ef of 10. The flat orange segment from ef 10 to ef 100 is the clamp.

3.6 Full-size base sets

At full size the HNSW default loses more recall: SIFT falls from 0.731 at 200,000 vectors to 0.696 at 1,000,000, and GloVe-100 from 0.515 to 0.478. GloVe-100 at full size never reaches 0.95 within the ef values I swept, up to 512. The IVF-Flat default improves slightly (0.528 to 0.560 on SIFT, 0.501 to 0.535 on GloVe-100) only because its fixed 100 lists each hold five to six times more vectors, so one probe scans more of the collection, at four to six times the latency. The free gain survives at full size: a tuned IVF-Flat index reaches 0.827 on SIFT and 0.810 on GloVe-100 at the default's latency. The subsampled study was not flattering the defaults; if anything it was generous to HNSW.

dataset	family	default recall 200k vectors	default recall full set	best at same latency, full set
SIFT (1,000,000)	HNSW	0.731	0.696 ± 0.026	0.696
SIFT (1,000,000)	IVF-Flat	0.528	0.560 ± 0.034	0.827
GloVe-100 (1,183,514)	HNSW	0.515	0.478 ± 0.041	0.478
GloVe-100 (1,183,514)	IVF-Flat	0.501	0.535 ± 0.038	0.810

Table 6. Default recall at 200,000 vectors against the full collections.

3.7 Code and reproducibility

All of this period's work is in the repository alongside the first report's. Two reproducibility checks passed: recomputing the dataset characteristics reproduced the first report's values to four decimals, and rebuilding the SIFT default index single-threaded reproduced its recall of 0.73067 exactly. The report itself is generated from the result files by `report/build_pr2.py`, and `report/check_numbers.py` re-derives every number quoted in this text from the data.

```
src/heuristic.py  src/sweep.py (profiles pr1, dense, kpass, scale)
scripts/make_synthetic.py  scripts/run_pr2_experiments.sh  scripts/analyze_pr2.py
scripts/build_variance.py  scripts/default_ci.py
scripts/live_demo.py  scripts/predict_demo.py
```

Repository: github.com/chinedu-2002/ann-recall-study

4. Challenges

My own headline did not survive more data. The first report said no default reached recall 0.75. With six datasets that is false; two HNSW defaults reach 0.93. I have restated the finding as what the data supports: the IVF-Flat gap is universal, and the HNSW gap depends on the dataset, largely through LID.

Rebuilding the same index changes its recall. The NYTimes default measured 0.679 in the first report and 0.604 in this period's sweep, with identical parameters. Tracing it: single-threaded `hnswlib` builds are deterministic and reproduce 0.679 exactly, but they always insert vectors in id order. With shuffled insertion order, three single-threaded builds gave 0.643, 0.646 and 0.652, and four two-threaded builds gave between 0.590 and 0.634. Across ten builds the same configuration spans 0.590 to 0.679 depending only on how the index was built. That 0.089

spread is as large as the entire ± 0.044 interval from sampling queries, and the interval does not account for it at all. So the first report's NYTimes figure was a favorable draw, and intervals over queries alone understate the real uncertainty. SIFT is steadier but not immune, spanning 0.709 to 0.743 across sixteen rebuilds. Table 3 keeps the first report's single-threaded figures for comparability; every new measurement this period used two-threaded builds.

The estimator saturates. Levina-Bickel reads 22.8 for a true intrinsic dimension of 48, so my synthetic generator cannot reach the LID of real text and word embeddings, around 43 to 53. The high end of the heuristic is anchored by two real datasets only.

Collection size is tangled up with LID. The two easiest datasets are also the two smallest. Default recall falls with collection size, so some of what looks like an LID effect may be a size effect. The heuristic over-predicts both MNIST sets (ef 32 for Fashion-MNIST against an oracle of 12, and 24 for MNIST against 16), which is what a missing size term would do.

Coarse evaluation. With six held-out datasets, one dataset moves a hit rate by seventeen points, and predictions are rounded to a discrete grid. I report hit counts as fractions of six rather than percentages for that reason. Separately, FAISS warns that 60,000 vectors is too few to train 4,096 lists, so the IVF results at that setting on MNIST and Fashion-MNIST come from under-trained indexes.

5. Next Steps

For weeks 7 and 8 I plan five things. First, separate collection size from LID by sweeping SIFT and GloVe-100 at 50,000, 100,000, 200,000, 500,000 and 1,000,000 vectors, then add log collection size as a second feature and repeat the held-out evaluation. Second, add a hubness measure to test whether it explains the two GloVe misses. Third, make the synthetic generator reach an LID of 40 to 55, by using a rougher map or heavier noise, so the upper half of the fit is not resting on two points. Fourth, given the rebuild effect, build every oracle setting three times with shuffled insertion orders so the oracle itself carries an error bar. Fifth, extend the heuristic from query parameters to build parameters (HNSW M and IVF nlist), which is where most of the memory cost is decided. I will also start drafting the final report's methods section while the design is fresh.